MINGWEI XU

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ABOUT ME

My research interests lie in **LLM post-training**, **multimodal reasoning**, **agentic AI**, **world model** and **self-improvment**. My work spans model alignment and reasoning, multimodal test-time improvement and **AI for scientific discovery**. I aim to build reliable and adaptive AI systems that can reason over complex evidence.

EDUCATION EXPERIENCE

University of Washington 2025.09 – Present
M.S. in Electrical and Computer Engineering Seattle, WA, USA
• **GPA:** 3.8/4.0
• **Selected Coursework:** Algorithms and Data Structures, Machine Learning, AI for Robot Control

Zhejiang University of Technology 2021.09 – 2025.06
B.S. in Automation Hangzhou, Zhejiang, China
• **GPA:** 3.6/4.0; **Rank:** 3/118
• **Honors:** Outstanding Graduate; Zhejiang Provincial Government Scholarship (3×); First-Class Outstanding Student Scholarship (2×); Innovation and Entrepreneurship Scholarship (2×)

INTERNSHIP EXPERIENCE

Zhejiang Lab 2025.07–2025.09, 2026.07–2026.09
Research Intern, LLM Post-Training and Scientific Discovery
• **LLM Post-Training:** Investigated post-training and alignment strategies for **Qwen-14B** and **32B** on an **8×A100 GPU** cluster; conducted reinforcement learning experiments (**PPO**, **GRPO**) with **LoRA** across model scales; designed **benchmark evaluations**, **ablation studies** and pipelines to analyze training and prompt.
• **GeoGPT & Scientific Discovery:** Conducted **pretraining** and **instruction tuning** of GeoGPT for geoscience reasoning; curated and optimized domain-specific training datasets for post-training.
• Developed a **scientific Agent** for automated hypothesis generation and research in protein–disease interactions, integrating **RAG**, evidence retrieval and **PPI networks** for biomarker discovery, validation and interpretation.

Hangzhou Closeli Technology Co., Ltd. 2024.05–2024.07
Automation Software Engineer Intern, Embodied Robot Control
• Developed embedded control software for a multi-joint robotic arm on **STM32**, implementing real-time sensor communication (**UART or SPI or I2C**), trajectory planning and **closed-loop PID** control modules.
• Designed a sensor-feedback pipeline integrating **force or torque** and **proximity signals** for adaptive motion execution, enabling real-time grasping force adjustment and obstacle avoidance.
• Collaborated on testing, debugging latency and **control loop stability** across multi-sensor configurations.

RESEARCH EXPERIENCE

Plug-and-Play Verifier for Agentic Reasoning [Github] 2026.06-2026.09
• Proposed a **plug-and-play verifier framework decoupling process verification** from policy optimization, integrating **domain-specific verifiers** for retrieved evidence, code execution, visual crops, and mathematical equalities to produce **bounded, step-attributed process rewards** through a unified interface; designed a **post-training schedule** with outcome-only warmup and group-shared process-outcome reward.
• Derived a **last-mile learning signal** that distinguishes trajectories with identical outcome rewards through differences in verified process returns; introduced **verification-privileged self-consistency** and explored **verifier-induced self-reflection** to encourage reliable intermediate reasoning and error correction
• Evaluated **Qwen3.5-4B** and **Qwen3.5-9B** across **AIME 2025**, **HotpotQA**, **LiveCodeBench**, and **MATH-Vision**, improving average accuracy over GRPO from **54.71% to 57.82%** and **61.58% to 64.16%**.
• Paper is under review at **ICLR 2027**.

Positive-Only Policy Optimization for LLM Reasoning (POPO) [Github] 2025.11–2026.09
• Existing **RLVR** relies on negative updates that destabilize training under sparse binary rewards. We proposed **POPO**, a positive-only framework eliminating negative updates while maintaining stable on-policy optimization.
• Designed a **self-distillation** architecture combining bounded importance weighting, EMA anchoring, representation-space alignment and entropy regularization, preventing reward hacking and collapse without negative rollouts.
• Evaluated POPO on **Qwen2.5-Math-1.5B** and **7B**, **DeepSeek-R1-Distill-1.5B** and **7B**, Llama-3.1-8B and

DeepSeek-Math-7B, achieving **36.67% Pass@8** on AIME 2025 with Qwen2.5-Math-7B, vs. **30.00%** with GRPO.
• Paper is accepted to the **ICONIP 2026** and **ICML Workshop DEMO 2026**.

AV-Phys Bench: Physics in Joint Audio-Video Generation [**Website**] 2026.03–2026.09
• No existing **benchmark** evaluates whether **joint audio-video models** respect **physical commonsense**. We contributed to designing a **physics-grounded** evaluation framework addressing this gap.
• Developed the **benchmark** with **321 prompts** across Steady State, Event Transition and Environment Transition, with **Anti-Physics** stress tests and fine-grained rubrics for visual, audio and cross-modal consistency.
• Built **AV-Phys Agent**, a **ReAct-style MLLM** evaluator integrating deterministic acoustic tools to automate rubric-based assessment, achieving rankings **aligned with human evaluation** across seven generation models.
• Paper is accepted to the **NeurIPS 2026**.

Knowledge-Informed Disease Protein Biomarker Discovery 2026.07–2026.09
• Biomarker selection optimizes prediction accuracy, ignoring biological relevance. We developed a **knowledge-informed framework** that balances predictive power with **protein–protein interaction (PPI)** evidence.
• Proposed a **PPI-Pareto multi-objective genetic algorithm** with **dynamic backward pruning** to optimize AUC, PPI quality, evaluation stability, and feature sparsity under a **predictive non-inferiority constraint**.
• Evaluated in **18 experiments** on Parkinson's (47,572 participants) and T2D (46,472 participants), improving **PPI quality** and **holdout AUC** across all six comparisons while reducing biomarker panels by **12–17%**.

Databricks Grounded Reasoning Cup (Google DeepMind Sponsored) [**Website**] 2026.02–2026.06
• Designed a **grounded reasoning Agent** with the **Google Antigravity SDK**, using deterministic two-tier routing to classify query scope, external-data requirements, temporal range and output format, then estimate complexity across lookup, aggregation or CAGR and statistical-analysis tasks to determine the reasoning strategy.
• Built a **grounding reasoning pipeline** that assembles **Gemini-embedding-2** and **FAISS** retrieval summaries, multimodal figure or table evidence, optional web and era-aware context, then selects tier-specific prompts and executes a single Agent with **Python** and **MCP corpus tools** for evidence-grounded multi-step reasoning.
• Represented the **University of Washington** in the Google track, using **Gemini models** across six rounds, competing against MIT, Yale, and Columbia on Databricks OfficeQA grounded reasoning over U.S. Treasury Bulletins (1939–2025); ranked **2nd** among the four Google-sponsored teams, outperforming MIT and Columbia.

SELECTED PUBLICATIONS

• Z. Cui, X. Liu, H. Fang, **M. Xu**, J. Liu, Z. Xu, W. Pian, S. Deng, F. Du, C. Ge, and Y. Tian, "Do Joint Audio-Video Generation Models Understand Physics?" *Advances in Neural Information Processing Systems (NeurIPS)*, 2026. (CCF-A; **Accepted**)
• **M. Xu** and H. Fang, "Improve Reasoning Ability by Reinforcing Only from Positive Rollouts," *ICML 2026 Workshop DEMO*, 2026. (**Accepted**)
• **M. Xu** and H. Fang, "Improving Reasoning Ability via Asynchronous On-Policy Self-Distillation under Positive Rollouts," *International Conference on Neural Information Processing (ICONIP)*, 2026. (CCF-C; **Accepted**)
• **M. Xu**, H. Deng, H. Xu, and Y. Zhang, "Time Series Forecasting and Correlation Analysis of the New Energy Electric Vehicle Industry Using ARIMA Algorithm and LSTM Neural Networks," *International Conference on Data Analytics, Computing and Artificial Intelligence (ICDACAI)*, 2024. (EI; **Published**)
• J. Li, H. Yang, **M. Xu**, Y. Wu, X. Shou, Z. Huang, Y. Hao, et al., "Task-Dependent Cortical Oscillatory Dynamics in Functional Constipation," *Sensors*, 2026. (JCR Q2; IF: 3.5; **Published**)
• B. Tu, Z. Yu, S. Jin, Y. Zhang, **M. Xu**, M. Zhang, Y. Jiang, X. Zhang, J. Weng, and X. Han, "Do Large Language Models Suffer from Cognitive Overload? A Benchmark and Orchestration Framework," *International Conference on Intelligent Computing (ICIC)*, 2026. (CCF-C; **Accepted**)

HONORS AND AWARDS

• **Meritorious Winner**, Mathematical Contest / Interdisciplinary Contest in Modeling (MCM/ICM), 2025.
• **National First Prize & 2nd Place**, National Undergraduate Intelligent Car Competition, 2024.
• **National Third Prize**, China Robot Competition and RoboCup China Open – Underwater Track, 2024.
• **National Third Prize**, National Undergraduate Electronics Design Contest, 2023.
• **National Second Prize**, National Undergraduate Mathematics Competition, 2022.

SPECIALTIES AND SKILLS

• **Programming language:** Python (Pytorch, Tensorflow, Sklearn, NumPy, Scipy), MATLAB, C++, C, Java
• **Tools:** Git, Linux, Altium Designer, FSDP, vLLM, Adobe Illustrator, Latex, Github, Microsoft Office
• **Language:** Mandarin (native), English (fluent)
• **Reviewer:** Conference on Language Modeling (**COLM 2026**); International Conference on Learning Representations (**ICLR 2027**); International Conference on Neural Information Processing (**ICONIP 2026**).